Waze Churn Project | Regression Modeling

Executive Summary

ISSUE / PROBLEM

After performing EDA and statistical analysis, we've been tasked to develop a regression model for identifying the correlation between different variables and the user churn status.

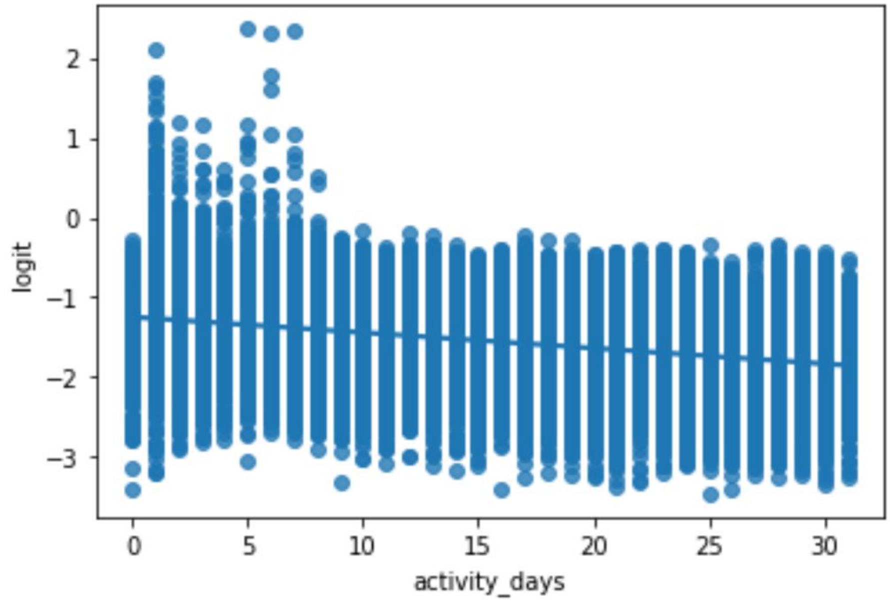
RESPONSE

At first step outliers are missing values were handled. Then feature engineering were done to develop and choose variables that have the most effect on churn status. In the construction, the assumptions of logistic regression model were validated. And data splitted to train and test the model. Then the model was developed.

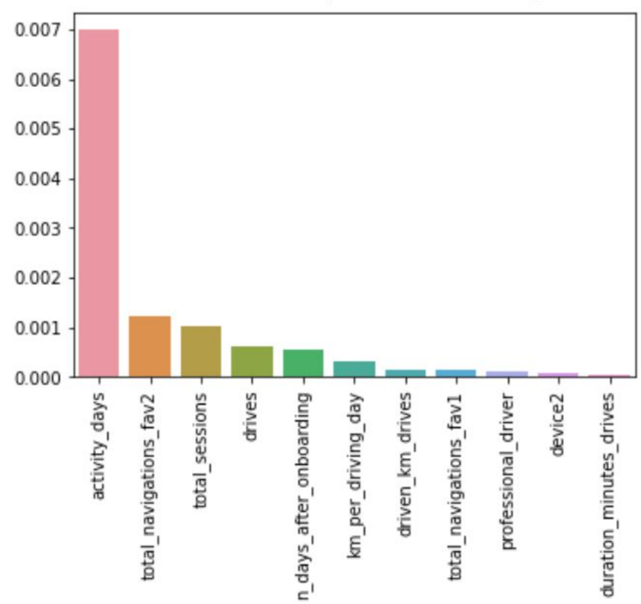
IMPACT

The model showed accuracy of 82.2%, Precision of 50% and recall of 3.3%. Although this shows good accuracy in predicting retained users, it's not as good as desired for churn prediction. Therefore the model is limited for use in further data exploration.

The Impact of activity days on logit (probability of user to churn)



Absolute of different variables' impact on probability of user to churn



KEY INSIGHTS

- Activity days had the most influence on churn state, where users with higher activity days tend to churn less which is not surprising.
- Km per driving day had surprisingly less effect on churn status taking the 6th rank between other variables.
- We recommend use the model to investigate relations between variables and discover data trends and not use it for prediction since it has a very low recall score.
- We believe that adding features such as app rate by the user, session duration and location of user can be of great benefit for the next step of developing complex ML model.